Year 10 Mathematics

Graphs 1

3.3 Straight-Line Graphs

Straight-Line Graphs

STARTER · 5 MIN

Date: _____ **Title:** Straight-Line Graphs

LEARNING OBJECTIVES

- Find the gradient and y-intercept from y = mx + c
- Rearrange equations into the form y = mx + c
- Find the equation of a straight line

STARTER *Answer in your book. No calculators.*

1. Find the gradient of the line through (0, 1) and (2, 7)
2. Make y the subject of $y - 3x = 5$
3. Work out $10 \div (-5)$
4. Find y when x = 0 for $y = 4x - 3$

Starter — Answers

STARTER

Tick or correct in green pen

1. $m = \frac{6}{2} = \mathbf{3}$

2. $y = \mathbf{3x + 5}$

3. $10 \div (-5) = \mathbf{-2}$

4. $y = 4 \times 0 - 3 = \mathbf{-3}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the gradient and y-intercept from y = mx + c	☐
SILVER	I can rearrange equations into the form y = mx + c	☐
GOLD	I can find the equation of a straight line	☐

Tick each box in the plenary when you can do it on your own.

Introduction — y = mx + c

INTRODUCTION · 3 MIN

Every straight line (except vertical lines) can be written as y = mx + c.

Gradient

m is the gradient (the number in front of x).

y = 3x + 2: m = 3

y-intercept

c is where the line crosses the y-axis.

y = 3x + 2: c = 2

Rearrange

Make y the subject first.

2y = 4x + 6 → y = 2x + 3

Parallel lines

Parallel lines have the same gradient.

y = 2x + 1 and y = 2x − 5

Examples 1 and 2 — Reading y = mx + c

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint y = mx + c m = gradient, c = y-intercept

Example 1

Write down the gradient and the y-intercept of

$y = 4x - 3$

Example 2

A line has gradient 2 and y-intercept 5.

Write down its equation.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Gradient m = 4

y-intercept c **= −3**

Tip: c includes its sign.

Example 2

m = 2

c = 5

y = 2x + 5

Tip: put m and c into y = mx + c.

Task 1

PRACTICE · 9 MIN

BRONZE

Write down the gradient (m) and the y-intercept (c)

Hint y = mx + c
m = gradient, c = y-intercept

1. $y = 3x + 7$

2. $y = -2x + 1$

3. $y = \frac{1}{2}x - 4$

4. $y = 5 - x$

5. Write the equation of the line with gradient −3 and y-intercept 2

★ Which line is parallel to $y = 2x + 1$? $y = 2x - 5$ or $y = 3x + 1$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $m = 3, \quad c = \mathbf{7}$

2 $m = -2, \quad c = \mathbf{1}$

3 $m = \frac{1}{2}, \quad c = \mathbf{-4}$

4 $m = -1, \quad c = \mathbf{5}$

5 $y = \mathbf{-3x + 2}$

★ $y = \mathbf{2x - 5}$

Examples 3 and 4 — Rearranging to y = mx + c

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint y = mx + c m = gradient, c = y-intercept

Example 3

Find the gradient and y-intercept of

$2y = 6x + 8$

Example 4

Find the gradient and y-intercept of

$3x + 2y = 12$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$y = 3x + 4 \quad (\div 2)$

$m = 3$

$c = 4$

Tip: divide every term by 2.

Example 4

$2y = -3x + 12 \quad (-3x)$

$y = -1.5x + 6 \quad (\div 2)$

$m = -1.5$

$c = 6$

Tip: get the y term on its own first.

Task 2

PRACTICE · 9 MIN

SILVER

Write each equation in the form y = mx + c

Hint y = mx + c m = gradient, c = y-intercept

1. $2y = 4x + 6$
2. $3y = 9x - 12$
3. $y - 4x = 1$
4. $x + y = 7$
5. $4x + 2y = 10$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $y = \mathbf{2x + 3}$

2. $y = \mathbf{3x - 4}$

3. $y = \mathbf{4x + 1}$

4. $y = \mathbf{-x + 7}$

5. $2y = -4x + 10, \quad y = \mathbf{-2x + 5}$

Examples 5 and 6 — Finding the Equation of a Line

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $m = \frac{y_2 - y_1}{x_2 - x_1}$ $y = mx + c$

Example 5

(3 marks)

Find the equation of the line through

(1, 5) and (3, 11)

Example 6

(3 marks)

Find the equation of the line parallel to

$y = 4x - 1$ that passes through (2, 3)

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$m = \frac{11 - 5}{3 - 1} = 3$$

$5 = 3 \times 1 + c$

$c = 2$

$y = 3x + 2$

Tip: find m, then substitute a point to find c.

Example 6

$m = 4$

$3 = 4 \times 2 + c$

$c = -5$

$y = 4x - 5$

Tip: parallel means the same gradient.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $m = \frac{y_2 - y_1}{x_2 - x_1}$ $y = mx + c$

1. Find the equation of the line through (2, 7) and (4, 11) (3 marks)

2. Find the equation of the line through (−1, 5) and (3, −3) (3 marks)

3. Find the equation of the line parallel to $y = 3x + 2$ through (1, 8) (2 marks)

4. Find the equation of the line parallel to $2y = x + 4$ through (4, 1) (3 marks)

5. Show that (5, 12) lies on the line through (1, 4) and (3, 8) (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $m = 2, \quad 7 = 4 + c, \quad c = 3, \quad y = \mathbf{2x + 3}$

2. $m = -2, \quad 5 = 2 + c, \quad c = 3, \quad y = \mathbf{-2x + 3}$

3. $m = 3, \quad 8 = 3 + c, \quad c = 5, \quad y = \mathbf{3x + 5}$

4. $m = \frac{1}{2}, \quad 1 = 2 + c, \quad c = -1, \quad y = \mathbf{\frac{1}{2}x - 1}$

5. $m = 2, \quad c = 2, \quad y = 2x + 2, \quad 2 \times 5 + 2 = \mathbf{12}$, **so it is on the line**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write down m and c for $y = 6x - 2$

SILVER Write $2y = 8x - 6$ in the form $y = mx + c$

GOLD Find the equation of the line through (2, 1) and (4, 9)

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $m = 6, \quad c = \ \mathbf{-2}$

SILVER $y = \ \mathbf{4x - 3}$

GOLD $m = 4, \quad 1 = 8 + c, \quad c = -7, \quad y = \ \mathbf{4x - 7}$

Exam Practice — Gradient and Intercept

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $y = mx + c$ m = gradient, c = y-intercept

Question 1

(2 marks)

A straight line has equation $y = 5 - 2x$

(a) Write down the gradient of the line. (1)

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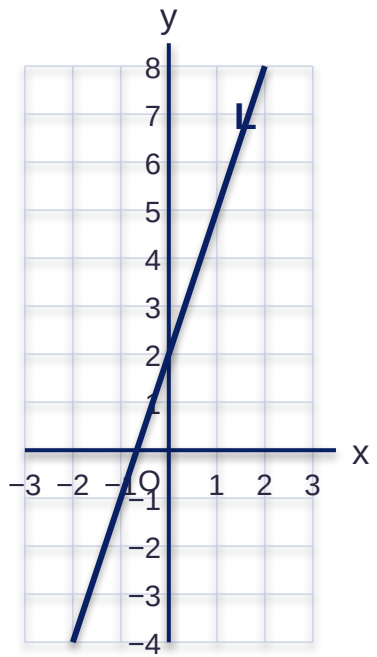
(b) Write down the coordinates of the point where the line crosses the y-axis. (1)

(..... , ...)

Question 2

(2 marks)

Find an equation of the line L.



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Exam Practice — Rearranging to y = mx + c

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint y = mx + c m = gradient, c = y-intercept

Question 3

(3 marks)

The line L has equation $4x + 2y = 10$

(a) Find the gradient of L. (2)

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(b) Write down the y-intercept of L. (1)

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Question 4

(2 marks)

Here are the equations of three lines.

A: $y = 3x + 1$

B: $2y = 6x - 5$

C: $y = 1 - 3x$

Which two lines are parallel?

Give a reason for your answer.

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Exam Practice — Finding Equations of Lines

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $m = \frac{y_2 - y_1}{x_2 - x_1}$ $y = mx + c$

Question 5

(3 marks)

Find an equation of the straight line that passes through the points (1, 4) and (3, 10).

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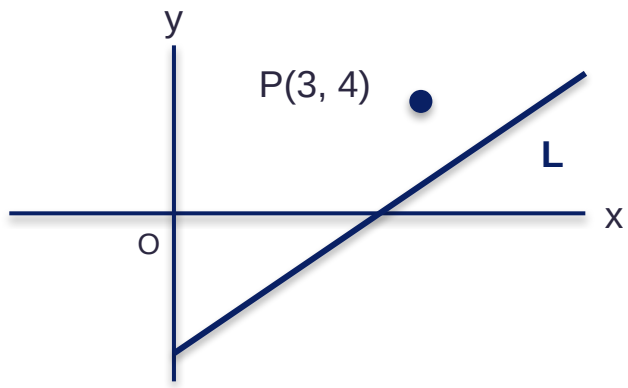
Question 6

(3 marks)

The line L has equation $y = 2x - 5$.

Find an equation of the line that is parallel to L and passes through the point P(3, 4).

Diagram NOT accurately drawn



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Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) −2 — B1

(b) (0, 5) — B1

Tip: y = mx + c. Here m = −2 and c = 5.

Question 2

Gradient = 6 ÷ 2 = 3 — M1

y = 3x + 2 — A1

Tip: read c where the line crosses the y-axis.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $2y = -4x + 10$ — M1

$y = -2x + 5$, gradient -2 — A1

(b) 5 — B1

Tip: make y the subject first.

Question 4

B: $y = 3x - 2.5$ — M1

A and B, both have gradient 3 — A1

Tip: parallel lines have equal gradients.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$m = (10 - 4) \div (3 - 1) = 3$ — M1

$4 = 3(1) + c \Rightarrow c = 1$ — M1

$y = 3x + 1$ — A1

Tip: find m first, then substitute a point.

Question 6

$m = 2$ — B1

$4 = 2(3) + c \Rightarrow c = -2$ — M1

$y = 2x - 2$ — A1

Tip: parallel ⇒ same gradient.